

Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

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Forename(s)

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Candidate signature

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INTERNATIONAL GCSE BIOLOGY

Paper 2

Monday 11 November 2019

07:00 GMT

Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a ruler with millimetre measurements
- a scientific calculator.

Instructions

- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice

In all calculations, show clearly how you work out your answer.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
TOTAL	



Answer **all** questions in the spaces provided.

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outside the
box

0 1

This question is about the structure of organisms.

0 1 . 1

Write the following parts of an organism in size order with the smallest first.

Use the words from the box.

[2 marks]

cell	organ	tissue
------	-------	--------

smallest

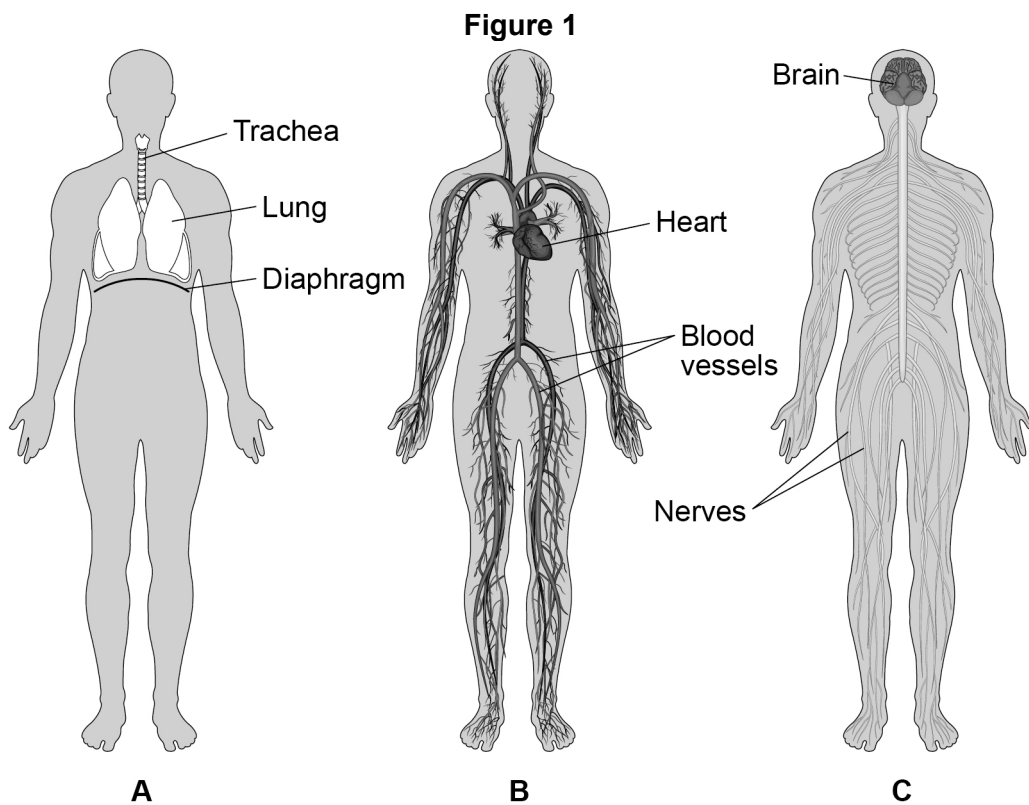
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largest

0 1 . 2

The human body has many organ systems.

Figure 1 shows three organ systems.



Complete **Table 1** to show the name of each organ system.

[3 marks]

Table 1

Organ system	Name
A	
B	
C	

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outside the
box

0 1 . 3 Draw **one** line from each organ system to its correct function.

[3 marks]

Organ system

Function

A

allows the body to move and
maintain its posture

breaks down food so it can be
absorbed

B

delivers oxygen and nutrients to
every part of the body

enables the body to react to its
surroundings

C

exchanges gases with the
environment

8

Turn over for the next question

Turn over ►

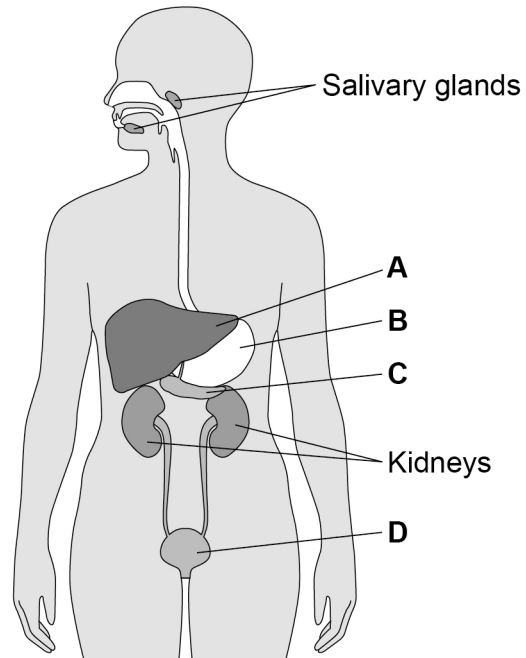


0 2

This question is about the liver.

Figure 2 shows some organs of the human body.

Figure 2



0 2

. 1

Which organ is the liver?

[1 mark]

Tick (✓) **one** box.

A

☐

B

☐

C

☐

D

☐


0 2 . 2 A man has a total body mass of 70 kg and a liver mass of 1.4 kg

What percentage of the total body mass is the liver?

[1 mark]

Tick (✓) **one** box.

0.1

☐

2

☐

5

☐

25

☐

0 2 . 3 The liver has many functions.

Complete the sentences below.

Choose answers from the box.

[2 marks]

gall bladder

iron

pancreas

fat

small intestine

urea

The liver produces bile which is stored in the _____.

The liver converts excess amino acids into _____.

Question 2 continues on the next page

Turn over ►



A patient with a diseased liver can have it replaced with part of a liver from a living donor.

After 2 months the livers of both the living donor and the patient are of normal size and can function well.

The use of living donors allows a donor who is closely related to the patient to be used.

0 2 . 4 Explain the advantage of using a donor who is closely related to the patient.

[2 marks]

0 2 . 5 Give **one** other advantage of using a living donor rather than a donor who has died.

[1 mark]

0 2 . 6 Living donors can also donate a whole kidney.

Living donors can still live a normal healthy life after donating a kidney.

Suggest why.

[1 mark]



The survival rate for patients receiving a transplant depends on several factors.

These include:

- the type of organ being transplanted
- the time from the transplant.

Table 2 shows the percentage survival rate from 1 to 10 years for patients receiving an organ transplant.

Table 2

Organ	Percentage (%) survival rate after transplant		
	1 year	5 years	10 years
Heart	79	66	57
Kidney	98	86	74
Liver	85	67	56
Lung	74	43	25

0 2 . 7

Which organ has the greatest survival rate 10 years after transplantation?

[1 mark]

0 2 . 8

What conclusion can be made about the relationship between the time from transplantation and the percentage survival rate?

[1 mark]

10

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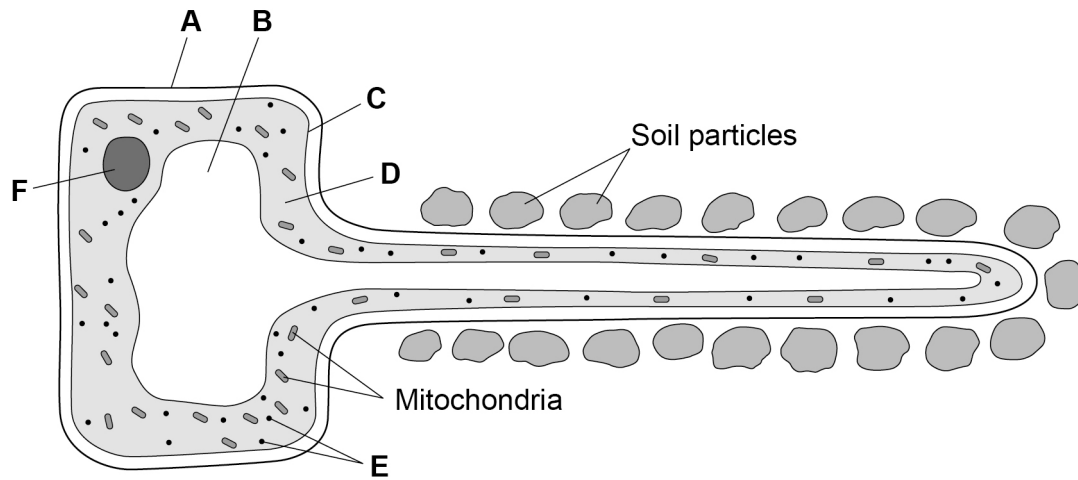
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0 3

Figure 3 shows a root hair cell of a plant.

Figure 3



0 3

1

Name the parts of the cell labelled **A** and **B**.

[2 marks]

A _____

B _____

0 3

2

Which part of the cell controls the movement of substances into and out of the cell?

[1 mark]

Tick (✓) **one** box.

C

☐

D

☐

E

☐

F

☐

[1 mark]

C

D

E

11

F

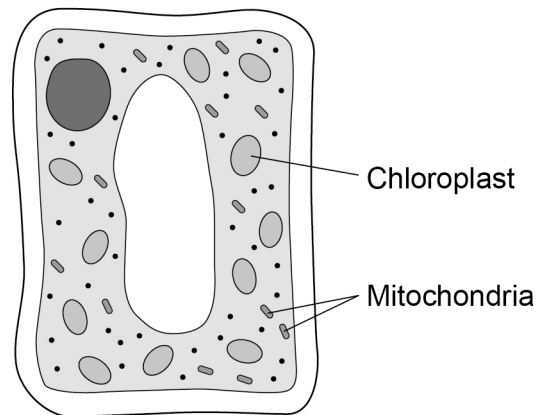
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[6 marks]

[illegible]

Figure 4 shows a leaf cell.

Figure 4



0 3 . 5 The leaf cell contains many chloroplasts.

Explain why the root hair cell does **not** need chloroplasts.

[2 marks]

0 3 . 6 Give **one** other difference between the leaf cell shown in **Figure 4** and the root hair cell shown in **Figure 3**.

[1 mark]



0 3 . 7 Leaf cells were placed into a solution. The solution was hypotonic to the cell contents.

Table 3 shows some possible results.

Table 3

	Movement of water	Effect on the leaf cells
A	into the cells	plasmolysis occurs
B	into the cells	turgor increased
C	out of the cells	turgor increased
D	out of the cells	plasmolysis occurs

Which letter shows the correct result?

[1 mark]

Tick (✓) **one** box.

A	☐
B	☐
C	☐
D	☐

0 3 . 8 Red blood cells were placed in water.

Describe what would happen to the red blood cells.

[1 mark]

0 3 . 9 Give the reason why the effect on red blood cells is different to the effect on leaf cells.

[1 mark]



0 4

A scientist compared the effect of three diets designed for athletes training for a marathon. A marathon is a 26-mile run.

This is the method used.

1. Separate 18 athletes into 3 groups.
2. The athletes eat different diets for a month:
 - athletes in group **A** eat a diet containing 30% carbohydrate
 - athletes in group **B** eat a diet containing 50% carbohydrate
 - athletes in group **C** eat a diet containing 70% carbohydrate.
3. Measure the concentration of glycogen in the muscles of each athlete before and after running a marathon.

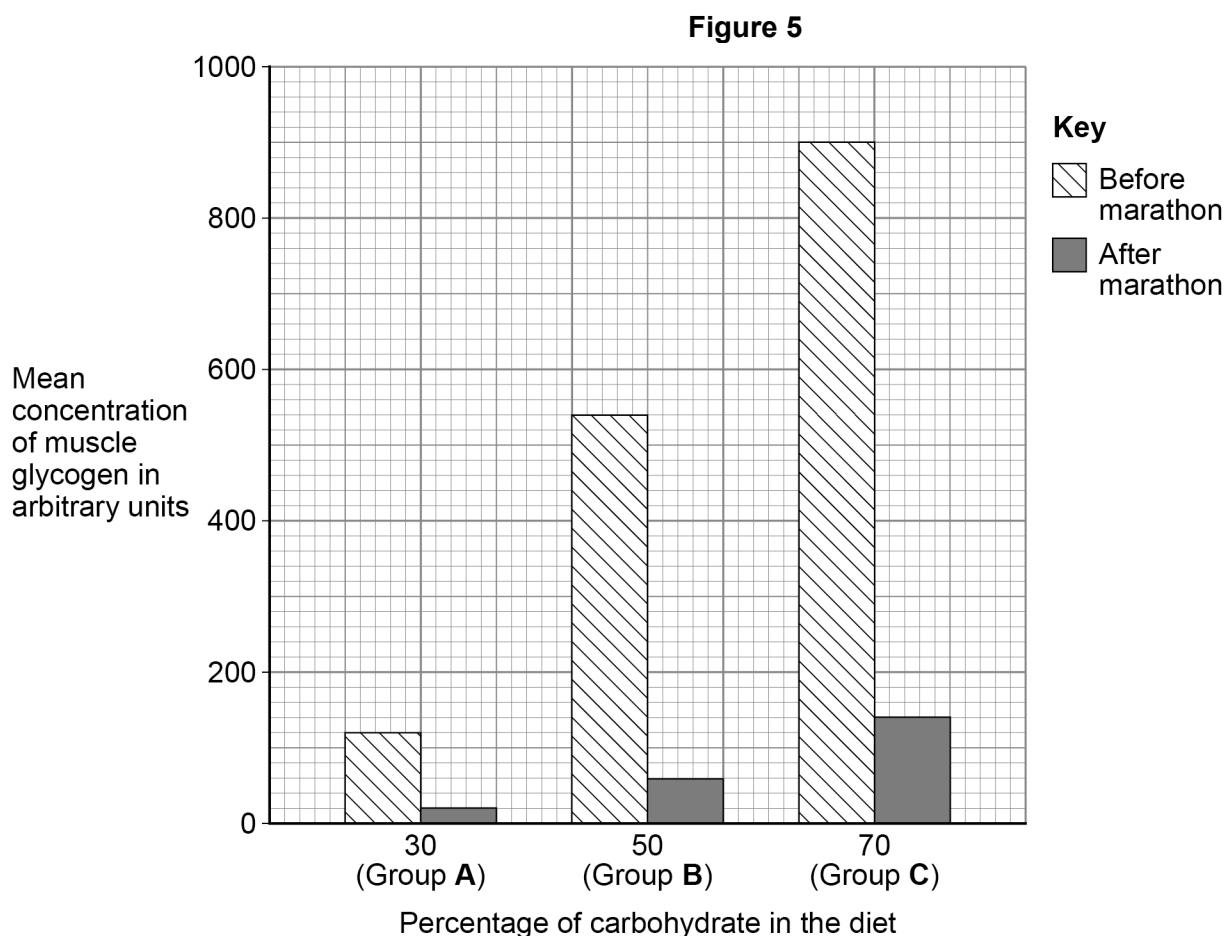
0 4

. 1

Give a **hypothesis** for this investigation.

[2 marks]

Figure 5 shows the results.



0 4 . 2

There were a number of control variables in the investigation.

Which **two** were control variables?

[2 marks]

Tick (✓) **two** boxes.

amount of food eaten per day

☐

concentration of glycogen in the muscles

☐

distance of the marathon

☐

length of time the diet was eaten

☐

percentage of carbohydrate in the diets

☐

0 4 . 3

Before the marathon, the concentration of glycogen in the muscles of athletes in group **C** was greater than in group **B**.

Calculate how many times greater.

Give your answer to **two** significant figures.

[3 marks]

Answer = _____

Question 4 continues on the next page

Turn over ►



0	4	.	4
---	---	---	---

The scientist recommended that athletes training for a marathon should eat a diet containing 70% carbohydrate.

Explain why eating this type of diet is an advantage.

[4 marks]

0	4	.	5
---	---	---	---

Explain why the muscles of athletes become fatigued by the end of a long run.

[3 marks]



0 4 . 6

After exercise the breathing rate **and** heart rate of an athlete remain high for several minutes.

Explain why.

[4 marks]

0 4 . 7

After doing a similar exercise, the breathing rate and heart rate return to normal more quickly in an athlete than in a person who does **not** exercise regularly.

Suggest why.

[1 mark]

19

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0 5

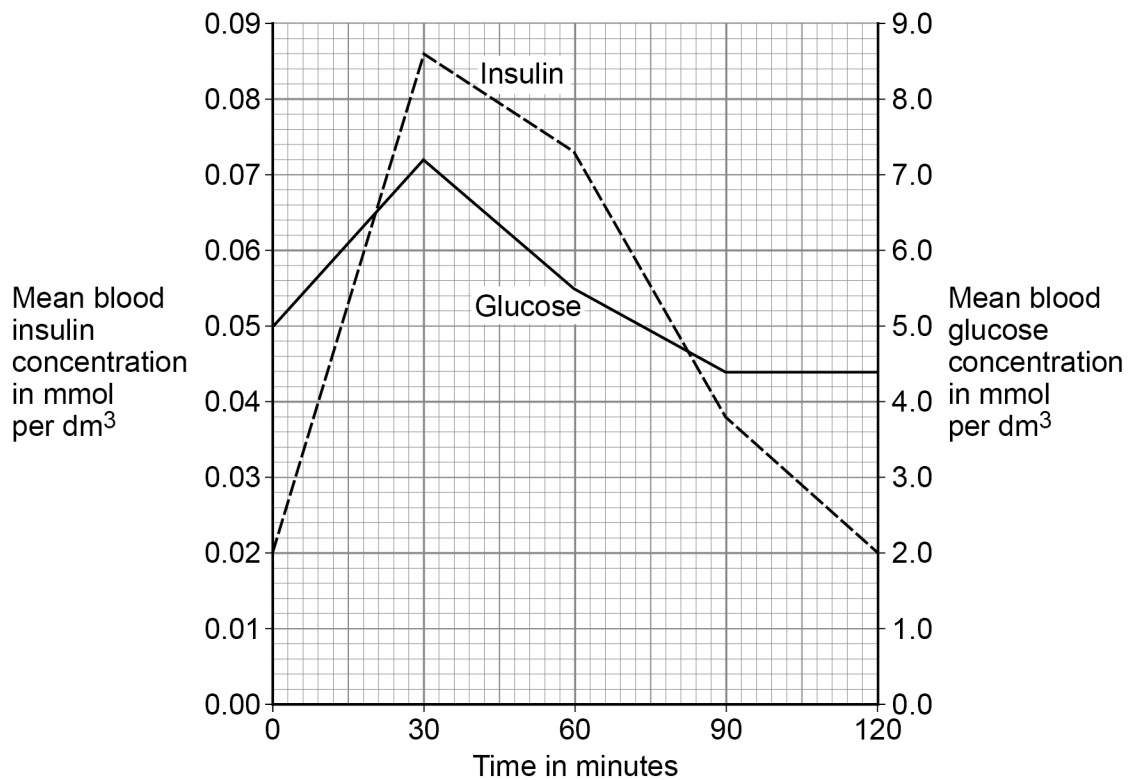
Scientists investigated the relationship between the concentrations of insulin and glucose in the blood of healthy people.

This is the method used.

1. Take a blood sample from each of 50 volunteers.
2. Each volunteer drinks a solution containing 50 g of glucose.
3. Take blood samples from each volunteer every 30 minutes for 2 hours.
4. Measure the concentrations of insulin and of glucose in each sample.
5. Calculate the mean values for blood insulin concentration and for blood glucose concentration for the volunteers at each time interval.

Figure 6 shows the results.

Figure 6

**0 5 . 1**

Calculate the rate of increase in the mean blood insulin concentration from 0 to 30 minutes.

[2 marks]

Rate of increase = _____ mmol per dm³/minute



0	5	.	2
---	---	---	---

Describe the changes in the mean blood glucose concentration during the investigation.

Use data from **Figure 6**.

[3 marks]

0	5	.	3
---	---	---	---

Explain the changes in the mean blood glucose concentration during the investigation.

[3 marks]

Question 5 continues on the next page

Turn over ►



0 5 . 4

The volunteers must remain seated throughout the test.

Explain why this is important.

[2 marks]

0 5 . 5

The blood samples were taken from capillaries.

Explain why blood glucose concentration is lower in veins than in capillaries.

[3 marks]

0 5 . 6

Type 1 diabetes is a condition which leads to a high blood glucose concentration.
Part of the treatment for Type 1 diabetes is a carefully controlled diet.

Suggest why a person with Type 1 diabetes should eat foods containing starch rather than glucose.

[2 marks]



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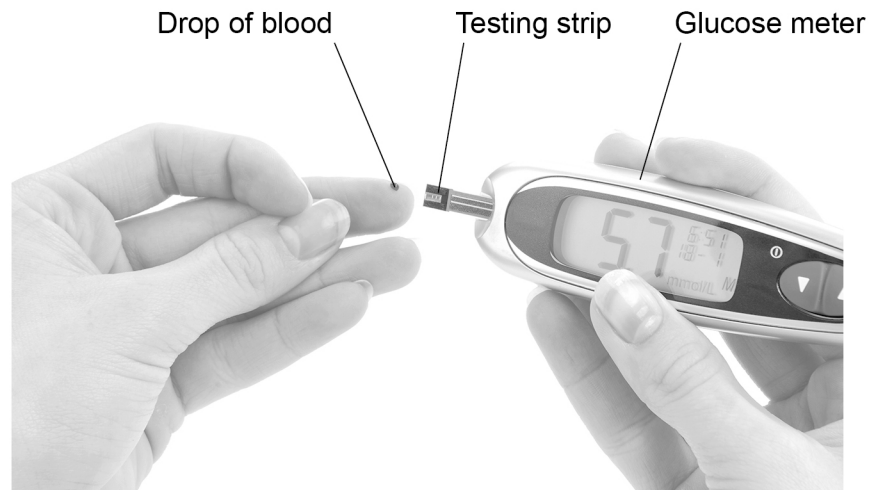
0 5 . 7 People with diabetes monitor their blood glucose concentration.

One way to monitor blood glucose concentration is to use the finger-prick test. In the finger-prick test, a drop of blood is inserted into a glucose meter. The glucose meter displays the result.

The test must be done 4 to 10 times every day.

Figure 7 shows the finger-prick test.

Figure 7



Another way to monitor blood glucose concentration is to use the sensor method.

A sensor is attached to the arm. Part of the sensor is inserted beneath the skin. The sensor stays in place for 14 days and automatically takes readings every minute. The results are transmitted to a reader and stored.

Figure 8 shows a sensor and a reader.

Figure 8



- less time was spent with either a high or a low blood glucose concentration
- 1% found the pain unacceptable during attachment of the sensor
- 8% complained of irritation in the skin around the sensor.

[5 marks]

[illegible]

20

Turn over for the next question

Turn over ►



0 6

A factory polluted a river by releasing a toxic chemical called mercury into the water. Mercury is absorbed and attaches to proteins in the tissues of organisms.

Figure 9 shows a food chain in the polluted river.

Figure 9

Microscopic algae → Insects → Small fish → Large fish

0 6**. 1**

The river water contained a low concentration of mercury.

Explain why high concentrations of mercury were found in the large fish.

[3 marks]



The level of mercury pollution was monitored by counting the number of salmon fish found in the river each year.

Salmon fish are more commonly found in water with lower levels of pollution.

Table 4 shows some of the results.

Table 4

Year of monitoring	Number of salmon fish	Level of mercury pollution
1990	1	Very high
1995	28	Very high
2000	722	Low
2005	363	High
2010	558	Moderate
2015	708	Low

0 6 . 2

Give **three** reasons why the method can only give an indication of the concentrations of mercury polluting the river water.

[3 marks]

- 1 _____

- 2 _____

- 3 _____

Question 6 continues on the next page

Turn over ►



Scientists investigated the concentration of mercury in two groups of 70 people.

- one group had eaten fish from the polluted river
- the other group had eaten fish from an unpolluted river
- the concentration of mercury in parts per million (ppm) was measured in blood samples taken from the people in both groups.

Table 5

	Source of fish	
	Polluted river	Unpolluted river
Range of mass of fish eaten per month in g	100–5250	1–4000
Mean mass of fish eaten per month in g	1290	340
Range of mercury concentration in blood in ppm	0.2–17.8	0.5–3.2
Mean mercury concentration in blood in ppm	3.43	0.90

Eating fish from the polluted river increased the concentration of mercury in the blood.

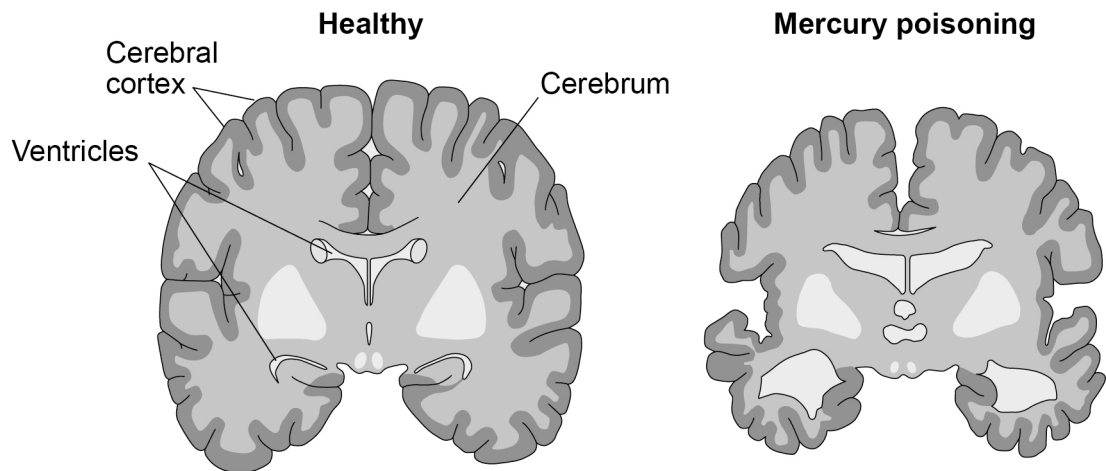
Use data from **Table 5**.

[4 marks]

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0 6 . 4 Figure 10 shows the scans of two brains.

Figure 10



Describe the effect of mercury poisoning on the brain as seen in the scans.

[3 marks]

Question 6 continues on the next page

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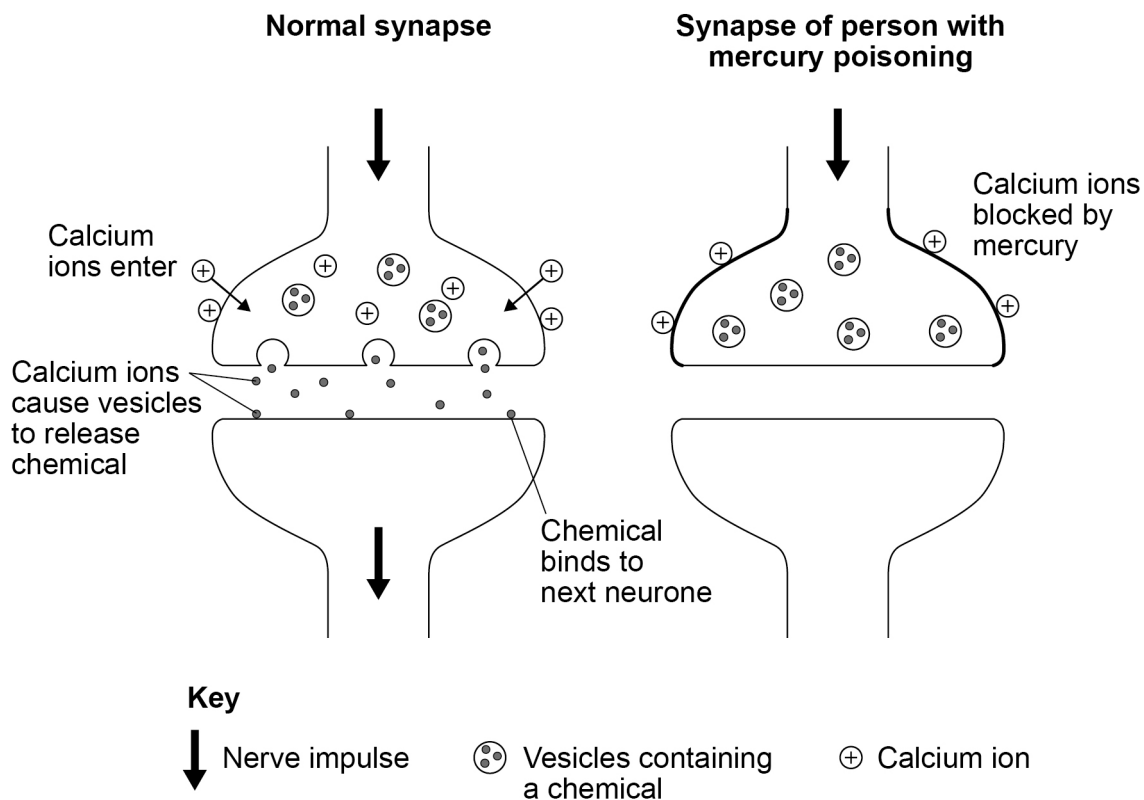


0 6 . 5

The effects from long-term exposure to mercury include blurred vision.

Figure 11 shows how chemicals are normally released at a synapse and the effects of mercury on chemical transmission.

Figure 11



Explain how the changes shown in **Figure 11** could lead to blurred vision.

You should include details of the nervous pathway.

[4 marks]



END OF QUESTIONS



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